



Sand ecologies, livelihoods and governance in Asia: A systematic scoping review

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ABSTRACT

Sand, gravel, and crushed rock – known as construction aggregates – are in high demand in the Asian region. Such demand is driven by high rates of urbanization, infrastructure development, and dam building: an unprecedented amount of sand is being extracted from the region's river, delta and estuary areas, only to be transported for infill or construction purposes elsewhere. This systematic scoping review examines the state of knowledge in the peer review literature on sand ecologies, livelihoods, and governance in Asia. We find that the literature mainly focuses on the ecological implications of sand mining, namely biotic and abiotic components: sand mining is linked with many forms of ecological degradation, although partial ecosystemic recovery may be possible when sand mining stops. In contrast, the limited analysis on livelihoods suggests that violence, work-related injuries, and precarious jobs are common for those working in the sand industry, with sand mining producing different types of work depending on the level of mechanization. We conclude by noting several gaps in the literature, including the narrow geographical focus (mainly India and China), the lack of attention to the intersection between sand mining and other anthropogenic disturbances, and the need to establish transparent sand governance processes within this region.

1. Introduction

Sand, gravel, and crushed rock – known as construction aggregates – constitute the largest share of anthropogenic mass globally (Torres et al., 2021), with Asia being at the center of the sand exploitation phenomena (Larson, 2018). Sand demand is fuelled by urbanization (Hugo, 2019) and infrastructure development (Torres et al., 2021), including massive investments in dams (Hackney et al., 2020). This is linked to the ubiquitous nature of sand: sand is used in concrete, glass, electronics, cosmetics, and pharmaceuticals, among other usages (Bendixon et al., 2021). Meanwhile, the global urban population is expected to grow from 4.3 billion people today to 5.6 billion in 2035, with significant urban growth taking place in Asian cities (UN-Habitat, 2021). Asian cities face an acute need for infrastructure development, and much of this development relies on sand. Such demand will further intensify as countries address climate change challenges, in particular sea level rise: vast volumes of sand are needed for land reclamation and flood protection (Torres et al., 2021). This is all happening at the very moment when sand scarcity is appearing as a global crisis (Bendixon et al., 2019), particularly in the Asian region that we will focus on (Torres et al.,

2021).

The 2019 sand governance report, produced by the United Nations Environment Program (UNEP, 2019), notes both a lack of governance - there is no global observatory for sand - and research in relation to sand. To be sure, scholarly attention to sand has increased in the past few years, moving beyond synthetic articles in *Science* or *Nature* to begin unpacking sand in its granular form. Geographers have been at the forefront of such conceptualization, as seen in papers that examine the materiality, temporalities and timescales of sand (Kothari and Arnali, 2020; Torres et al., 2021), reflecting on the role of sand in the built form (John, 2021; Moser and Côté-Roy, 2021), and considering sand as a granular system (Jamieson, 2021a). This has led to compelling debates about how to think about sand, as seen in the recent exchanges on the concept of granular geography in the *Dialogues in Human Geography* (Jamieson, 2021b; Kothari, 2021). Further social science thinking placing sand into conversation with illicit supply networks (Magliocca et al., 2021), flow studies (Lamb and Fung, 2022) and precarious labour regimes (Marschke et al., 2021) shed additional insights.

Even so, sand remains an understudied commodity, with journalists (e.g., Beiser, 2018) and activist groups (e.g., Mother Nature, Sand

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Stories, Asienhaus) playing a significant role in highlighting particular cases, including deaths associated with the criminal nature of sand mining in India (Mahadevan, 2019), rampant sand mining in Bangladesh (Vice News, 2021), changes to China's Poyang Lake linked to decades of sand mining (Hernandez et al., 2021), and Singapore's purchase of sand from poorer regional neighbours for infill and construction (Global Witness, 2010). These specific cases illustrate how sand flows stretch well beyond national borders, as sand moves through rivers into water basins and delta estuaries before being dredged, displaced, and transformed into a material form. Sand circulates before it becomes fixed across instable commodity frontiers (Jamieson, 2021a). These flows pose serious challenges for the Asian region as sand is depleted, altering ecosystems at sites of extraction and sites of production, while further making changes to sand and non-sand livelihoods across multiple sites (Marschke et al., 2021).

In this paper, we examine the English-language peer-review literature that focuses on the social-ecological aspects of sand demand in Asia. Our aim is to understand what the scholarship tells us about sand mining across Asia, including the ecological challenges emerging through sand mining, but also how sand extraction impacts people's livelihoods and their working conditions. Our systematic scoping review reveals a robust ecological literature on the impacts of sand mining, particularly in terms of geomorphological change and species decline seen across key Asian rivers and deltas. This literature is far richer than that focusing on sand livelihoods, working conditions or sand governance. Sand remains a cheap, and at times illicit, commodity that supports Asia's development: there is a cost to such rapid, chaotic development, including social tension, sand depletion, and overall social-ecological decline. There is a need to further connect the ecological impacts of sand mining with broader anthropomorphic change, and to unpack current barriers for sand governance.

2. Methods

The primary approach used to gather studies relevant for inclusion in this review was database searching across the peer-review literature. This process involved developing and testing search strings in four databases – ProQuest, Scopus, Environmental Impact, and Web of Science

– to retrieve references. Initial keywords were identified based on a preliminary review of the literature conducted by the research team, from which the final search strategy was developed and then applied across all four databases. Search strings were designed to accommodate various combinations of three major themes within Asia's sand mining sector: (a) ecological implications, (b) livelihood issues and (c) governance challenges. Searching was conducted in March 2019.

Relevant bibliographic data were exported from each database and imported into the Covidence systematic review management software. A total of 1348 articles were identified using the approach of structured database searching, from which 438 duplicates were removed as part of the deduplication process, leaving 910 articles for initial screening (Fig. 1). Titles and abstract of those articles were screened by the first author and a research team collaborator. Our inclusion criteria included papers that address ecology, livelihoods or regulation within the sand industry, broad human-environment dimensions related to sand, or told something about the impacts of sand mining (particularly in terms of ecology or livelihoods). Our exclusion criteria included non-English publications, theses and dissertations, books, audio-visual media and grey literature; papers focusing on desert sand, desertification, sandstone, sand dune, sandalwood, oil and gas; and papers not focused on Asia. Any disagreements in screening were discussed in person to resolve conflicts, and a total of 261 articles passed the first round of screening.

The second round of analysis involved a full text screening of the 261 articles that had addressed one of the main themes (ecology, livelihoods, governance of sand mining) in their abstract. In this round of screening, we further excluded papers that were published pre-1995, only paid attention to sand in a fleeting manner within the text of the paper, or articles that were particularly difficult to access (such as publications from India or China that were not available online through library systems from different Canadian-based institutions). After the full text screening, a total of 129 studies were included for this review. We then turned to a basic quantitative questionnaire to scope and characterize these articles. Our analysis proceeded by coding and organizing content within the articles according to our major themes (ecology, livelihoods, governance).

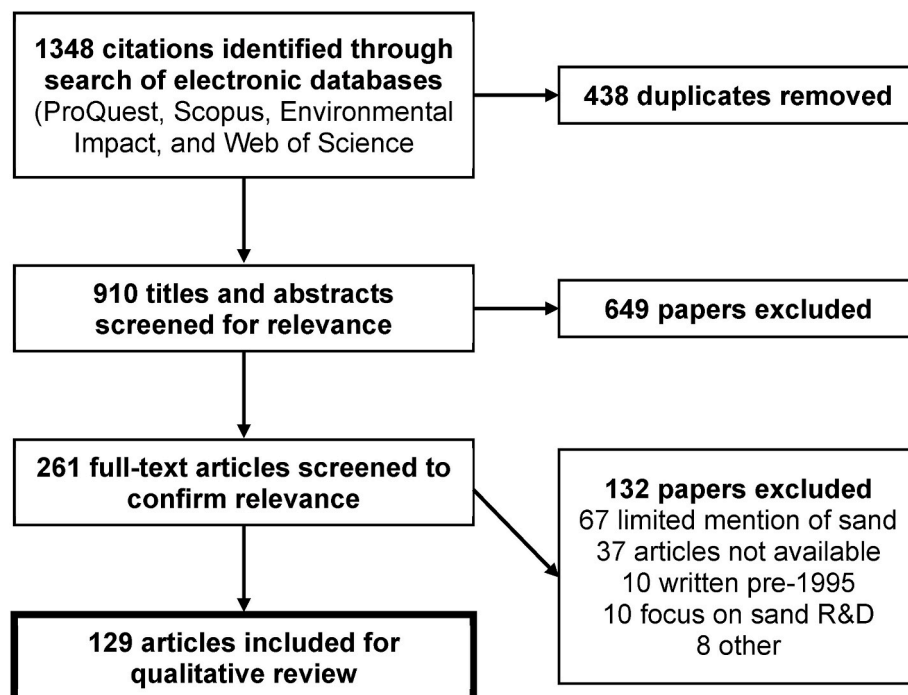


Fig. 1. Flow chart of the selection of studies.

3. Results

3.1. Key characteristics of the literature

The majority of the 129 articles we surveyed focused on China (58 articles) or India (38 articles) (Fig. 2). The remaining articles were case examples from other countries, including Cambodia, Indonesia, Iran, Korea, Oman, etc. Most lead authors (all but 11) were based at Asian institutions, mainly universities. Nearly all the articles (126) drew on original research, with the remaining articles being review articles (2) or offering a commentary (1). While most of the literature we reviewed was focused on ecology (102), a few articles did focus on livelihoods (27), including livelihood opportunities and exclusions emerging from sand mining and broader economic development. Insights on sand governance was sparse across the literature, limited to general recommendations, critiques or best practice examples.

3.2. Understanding the ecological implications of sand mining

The bulk of the literature on sand mining in Asia focused on ecological impacts, particularly in China and, to a lesser extent, India. The most common theme was ecological change occurring in freshwater ecosystems, in part because such environments are the most suitable for the extraction of construction sand (Mujabar and Chandrasekar, 2012). This might also be linked to the visible nature of river sand mining in contrast to quarry, coastal or open water sand mining. Research also examined marine dredging activities (e.g., Cho, 2006; Kim, 2009), land-based sand mining (Kumar & Kumar 2014), brick factories that rely on nearby sand deposits (Bandyopadhyay et al., 2013), and navigational channel dredging (Rehitha et al., 2017).

The ecological impacts of sand mining span two broad categories, biotic and abiotic. Articles with a biotic focus encompass biomass and biodiversity change, including change and/or decline in vertebrate and invertebrate species as well as impacts on fauna. Articles that probe abiotic change, predominantly emphasise four aspects, including pollution generated by sand mining activities, hydrological modifications, geomorphological change, and an analysis of how these combine to drive water level change. We detail our categorisation in Table 1.

3.2.1. Biotic impacts of sand mining

The biotic consequences associated with sand mining activities span vegetal and faunal biomass and biodiversity change, with modifications affecting both vertebrate and invertebrate species. Habitat alteration or destruction, machinery noise, and shifts in salinity all stand out as important processes driving biomass change and disturbances. For example, in China's Lake Poyang, researchers have documented how sand dredging has damaged habitats where fish spawn and feed, particularly along migratory routes (Shao et al., 2012; Huang et al.,

Table 1

Ecological impacts of sand mining.

Category	Impact on	Example(s)
Biotic	Vertebrate species	Fish and mammal biomass and/or biodiversity decline
	Invertebrate species	Macrobenthic and plankton biomass and/or biodiversity decline
	Fauna	Decline of fauna and fungi
Abiotic	Pollution	Water pollution
	Hydrology	Air pollution Decrease in total sediment load
	Geomorphology	Increase in suspended sediment load Increased erosion
	Water levels	Bathymetric changes Increased or reduced risk of flooding
		Increased risk of drought

2013). In contrast, sand mining can also open opportunities for fishing, as dredging can drive geomorphologic change that enables terrain to be more accessible for fishing vessels and gear (Huber et al., 2015). Although changes in fish biomass can lead to some fish species thriving (Huang et al., 2013), the overall outcome is typically that of a declining fishery (Padmalal et al., 2008; Sheeba, 2009; Huang et al., 2013).

Sand mining impacts other vertebrate species too, including endangered aquatic mammals such as the Yangtze finless porpoise (*Neophocaena asiaeorientalis*), and is thought to have accelerated the extinction of the Yangtze dolphin (*Lipotes vexillifer*) (Chen, 2017; Larson, 2018). Sand mining also counts among anthropogenic activities that drive bird habitat change, with Aarif et al. (2014) linking the sand industry with bird population and species diversity declines in Kerala, India. Yet, such impacts are not permanent per se, with Shao et al. (2012) drawing a connection between the scaly-sided merganser (*Mergus squamatus*) population increasing after the termination of two sand-mining operations in China's Jiangxi Province.

Besides driving habitat destruction, sand mining increases water turbidity. Whereas we probe the abiotic impacts of this process in section 3.2.2, biotic consequences are related to less light penetrating the water. This in turn impels the photosynthetic activity from many aquatic species (Sheeba, 2009), including macroinvertebrate benthic species and phytoplankton. Research shows that sand mining leads to reductions in both the diversity and abundance of benthic species living on the water floor such as crustacean, shellfish and other macroinvertebrate species (Kaladharan and Vijayakumaran, 2012; Nelson et al. 2015, 2016; Rehitha et al., 2017; Meng et al., 2018; Mohamamad and Jalal, 2018). Kim (2009) estimates that the economic toll of such bottom species loss over a 4 km² area in Korea amounted to USD \$600,000 for local fisheries in 2001. The economic toll of sand mining on fisheries correlates with the duration and intensity of dredging activities undertaken in a specific site (Kim et al., 2008). Yet, Rehitha et al. (2017) conclude that benthic populations can recover after dredging has stopped, pending that sediment composition retrieves pre-sand-mining characteristics, a process that takes more or less time to occur depending on the scale of sand mining-driven disturbance.

Sand mining-driven water turbidity is also said to impact bacterial life, with Won et al. (2017) concluding that this activity reduces aquatic bacterial diversity in Korea. Comparing phytoplankton populations in Chinese rivers where sand is being mined versus populations from less disturbed waterways, Zhang et al. (2019) find that the former rivers are home to more heterogenous albeit less diverse and less abundant phytoplankton populations. Meng et al. (2018) concur, but also find that some benthic and phytoplankton species thrive in areas adjacent to dredging sites. The authors posit this might testify to these species being less sensitive to noise and vibrations than the predators feeding on them.

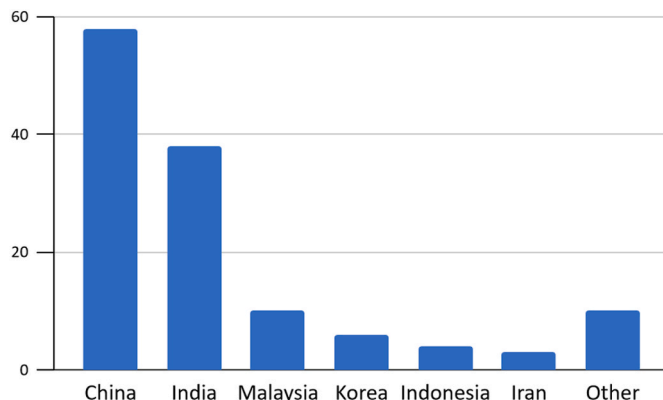


Fig. 2. Geographic scope of the literature (n = 129).

Sand mining likewise impacts fauna in both aquatic and terrestrial ecosystems. In Malaysia, [Hossain et al. \(2015\)](#) conclude that sand mining and the creation of sand depots cause seagrass loss, which furthers habitat loss for animal species. In south India, sand mining is one of many anthropological pressures that impact coastal sand dune health. For example, [Ghate et al. \(2014\)](#) found that fungi diversity is receding in such ecosystems. Heavy equipment piling together tonnes of aggregates can likewise destroy riparian vegetation ([Ashraf et al., 2011](#)). On-land mining in Rajasthan (India) leads to the systematic removal of vegetation, and areas recolonized by fauna after mines are decommissioned are typically not as biodiverse as they were pre-mining, as not all native species thrive in degraded sites ([Kumar & Kumar 2014](#)).

3.2.2. Abiotic impacts of sand mining

The literature we reviewed highlights that sand mining is an activity that yields both water and air pollution. In aquatic environments, dredging drives sediment swirling that then reshuffles deposited pollutants ([Ashraf et al., 2011](#)). Heavy rainfall results in higher phosphorous concentrations both nearby sand dredging sites and beyond ([Abidin et al., 2018](#)), which in turn fosters water eutrophication and further impacts water quality ([Tian et al., 2017](#)). Sand mining also influences dissolved organic carbon (DOC) concentrations, with [Xu et al. \(2015\)](#) finding that DOC levels in China's Poyang Lake are higher in areas where more intense dredging activities occur. The same process is hampering the capacity of estuarine and marine seabeds to sequester organic carbon in the Pearl River Delta region ([Lian et al., 2018](#)) and concurrently impels the capacity of this ecosystem to mitigate climate change.

Sediment swirling and barge overflow also influence heavy metal concentration levels, though in diverse ways. While the removal of sand and sediment leads to a decreased concentration in heavy metal loads, it simultaneously means more such pollutants are being eroded from bare rocks located along shorelines downstream from sand mining sites ([Prakash et al., 2016](#)). [Bayram and Önsoy \(2015\)](#) find that sand mining is associated with higher heavy metal levels in a Turkish river, a situation which, they suggest, could drive groundwater pollution too. Besides such potential groundwater pollution risk, sand mining is known to modify groundwater recharge, either positively or negatively ([Tejpal et al., 2014](#)). Dredgers are also associated with increased oil and wastewater pollution, as these pollutants are often pumped or dispersed in rivers directly in the course of sand mining operations ([Su et al., 2011](#)). The same vessels also cause air pollution through exhaust.

Sand dredging drives a series of interlinked hydrological changes. Research focusing on 'hydrologic alteration' addresses many of these changes together, highlighting a correlation between sand dredging and high degrees of hydrologic alteration in Chinese rivers ([Zhang et al., 2009b](#)). Specific hydrological changes include sand-mining driven variations on sediment budgets, which manifest in two main ways. First, the important quantities of aggregates being removed from Asian waterways lead to a decrease in the overall sediment load that rivers carry out to the sea, including the Yangtze and Pearl rivers in China ([Chen et al., 2006](#); [Chu et al., 2009](#); [Du et al., 2016](#); [Wu, C. et al., 2016](#)). Chinese rivers yield one tenth of global sediment discharging into oceans, with the global consequences of massive China-based sediment withdrawals remaining widely unknown ([Dai et al., 2009](#)). Sand mining also leads to the removal of coarser sediments, leaving the water floor covered with finer particles and impacting river channel flow ([Xu and Milliman, 2009](#); [Xia et al., 2013](#)). Sand mining influences sediment budgets in a second way, as dredging and barge overflow involve the stirring of important quantities of water and silt. This leads to resuspension of deposited sediments, driving a localised rise in suspended sediment concentrations and water turbidity ([Kim and Lim, 2009](#); [Feng et al., 2012](#); [Zheng et al., 2016](#); [Li et al., 2019](#)). Besides the impacts on flora and fauna, along with pollution impacts, increased turbidity is linked to water quality degradation and modifications in temperature gradients – that is the difference between surface and subsurface water temperatures ([Sharip and](#)

[Zaki, 2014](#); [Lusiangustin and Kusratmoko, 2017](#)).

Hydrological changes are closely related to geomorphological ones, and the actual removal of sand yields increased erosion – silt-free water being more erosive than silt-rich water – and decreased accretion, or decreases in the areas where sediments are being deposited ([Wu C. et al., 2016](#)). Sand mining-driven erosion has caused infrastructure damage, such as exposing bridge footings in Malaysia ([Julien et al., 2010](#)), and building collapse along the Ganges River ([Saha, 2016](#)). Such erosion has also led to interruptions in river transport along the Yangtze River ([Chen et al., 2006](#)). Sand mining is associated with further bathymetric (waterbed topography) changes such as waterway direction or thalweg shifts, as well as either the deepening, narrowing, or widening of river channels and deltas ([de Leeuw et al., 2010](#); [Kumar et al., 2010](#); [Zhang et al., 2011](#); [Mujabar and Chandrasekar, 2013](#); [Zheng et al., 2014](#); [Ghosh et al., 2016](#); [Liu et al., 2019](#)). Sand-mining driven mouth bar erosion is documented in the Pearl River Delta ([He et al., 2019](#)), as is the decrease in braided river areas along the Kerala coast ([Narayana and Priju, 2006](#)), and shoreline erosion and sand dune flattening in Tamil Nadu ([Mujabar & Chandrasekar, 2012, 2013](#); [Parthasarathy and Natesan, 2015](#)) and Kerala ([Prasad et al., 2016](#)). In some areas in West Kalimantan (Indonesia), sand mining and mangrove destruction lead to the shoreline receding an average of 20 m yearly ([Akbar et al., 2017](#)). Whereas geomorphological changes vary between sites, rising channel capacities stand out as a common outcome ([Zhang, W. et al., 2009](#)). These circumstances impact water velocity ([Zhang et al., 2015](#)) and monthly mean water discharges, with [Zhang Q. et al. \(2009a\)](#) concluding that sand mining affects high water levels to a greater extent than low water equivalents.

Similar to sea level rise and sinking coastal areas (e.g., subsidence), sand mining exacerbates tidal height and increases the tidal prism, which creates a greater risk for saltwater intrusion and makes coastal areas more vulnerable to typhoons ([Zhang and Hao, 2008](#); [Zhang Q. et al., 2010](#); [Zheng et al., 2014](#); [Chaussard and Kerosky, 2016](#); [Cai et al., 2019](#); [He et al., 2019](#)). In contrast, China-based research suggests that sand mining can reduce risks associated with upstream freshwater flooding ([Liu et al., 2014](#); [Li et al., 2015](#)), particularly when mining occurs in lakes ([Nakayama and Shankman, 2013](#)). For example, China's Poyang Lake now drains more quickly and holds a greater water storage capacity due to sand mining-induced bathymetric changes ([Li et al., 2015](#); [Mei et al., 2016](#); [Yao et al., 2018](#); [Ye et al., 2018](#)). Given this, water control infrastructure is now being considered to alleviate sand-mining related drought risk in that same lake ([Gao et al., 2014](#); [Lai et al., 2014](#); [Mei et al., 2015](#)). Lowered river levels can result in the use of mechanical pumps to supply freshwater to river-fed irrigation systems ([Julien et al., 2010](#)) and make water withdrawals in general more complicated ([Zhang et al., 2013](#)).

3.3. Understanding the livelihood implications of sand mining

Sand as a livelihood activity was not a predominant focus in the peer review literature. Around 20 percent of articles investigated aspects of sand livelihoods or broader economic development that can emerge from sand mining activities. Nearly half of these papers focused on India; the other articles were one-off cases spanning across Asia, from West Asia (Iran) to South Asia (Bangladesh), to Southeast Asia (Indonesia, Vietnam) and, finally, East Asia (China, Hong Kong). Attention to livelihoods was fleeting, with limited gender or class analysis (but see [Rege et al., 2016](#); [Lamb et al., 2019](#)). The few papers that focused on economic development linked uneven job creation in a context of environmental decline ([Singh & Kumar 2018](#)). Nonetheless, four themes emerged with regards to livelihoods and sand, namely: how sand provides work; how such work can result in health challenges; conflicts that can emerge with sand work; and how sand work impacts non-sand livelihood activities ([Table 2](#)).

The first category of papers focused on how sand provides work. This can be positive, particularly when sand work pays better than other

Table 2
Impact(s) of sand work on livelihoods.

Impact(s)	Example(s)
Provides work	Better pay than other similar low-skill jobs Generates economic activity in a region, including spin-off benefits Enhances worker precarity; may include violence & bonded labour
Work can result in worker health challenges	Workplace injury Ocular morbidity Malaria Respiratory disease Radioactivity exposure
Work can result in conflict	Heighten local inequalities Land grabs Damage infrastructure and/or ecosystems, exacerbating conflicts
Work can impact non-sand livelihoods	Fishers Farmers Tourist operators

similarly skilled jobs, reduces unemployment in a region, and helps to diversify income sources (Farahani and Bayazidi, 2018). Sand mining offers work to landless labourers and farmers in India, for example (Singh & Kumar 2018), or to Indonesian households living near sandy volcanic aggregate deposits (Gob et al., 2016). Sand livelihoods can be lucrative, particularly for those owning dredgers, boats, trucks or in management positions (Lamb et al., 2019). In other cases, however, authors cite how sand mining is linked to a lack of work alternatives (Rosyidie et al., 2018). For example, in places where running a home business (sewing, selling goods) or farming is in decline, sand mining may be one of the only viable low-skilled job alternatives. There are other cases where the presence of sand mining does not translate into jobs for local people, as seen in coastal Cambodia and parts of Myanmar (Lamb et al., 2019).

Sand work can also yield negative impacts, particularly when sand work leads to debt-bondage and exacerbates precarity (Singh and Tripathi, 2010). For example, since the early 2000s there has been evidence of child and bonded labour within sand work in India (Wazir, 2002), something that is further examined in the non-peer review literature (Mahadevan, 2019) and in a recent documentary on sand mining in Bangladesh (Vice News, 2021). Debt bondage is seen in rural areas and among India's Indigenous peoples, where hundreds of households remain under the control of particular sand mining contractors (Singh and Tripathi, 2010). Workers can be in debt to employers, with their children inheriting parental debt. Such debt bondage is further linked with the criminality found in India's sand industry (Rege, 2016).

India's sand mafias consist of a series of violent, impenetrable organized crime groups (Rege, 2016). Sand mafias are known to inflict harm to those who stand in their way, including intimidation, job transfers, physical violence, torture, and in some cases, murder or death (Rege and Lavorgna, 2017). Fear of violence is real: this in part explains why the corrupt system of sand extraction remains so pervasive. Villagers and local officials are complicit in sand mining activities, allowing sand to be stored on their property, falsifying documents, ignoring ecological destruction or staying quiet in cases of bonded labour practices, because a sand mafia strategically offers protection for those who support sand mining activities (Rege, 2016). Corruption is rampant across the police, construction and political sectors. Poor regulatory measures further exacerbate sand mining activities (Rege and Lavorgna, 2017). India's sand mafias focus on profit, with a core of workers (who may face debt bondage) working in the industry, and extra labour being drawn in as needed, including child labour, fishers and farm workers (Rege and Lavorgna, 2017).

A second category of papers focused on health challenges related to sand work. One study found that 30 percent of sand miners had at least one occupational health condition (Rajesh and Naik, 2016). Another

study noted that sand miners experienced a 93 percent prevalence rate of musculoskeletal pain and discomfort, ranging from lower back, shoulder and neck pain (Mohapatra, 2017). Skin and eye problems are another issue. Multiple studies found that sand workers are continuously exposed to sunlight and glare, which can damage eyes and cause a series of ophthalmic conditions including eye cataracts and dimness of vision (Rajesh and Naik, 2016; Mohapatra, 2017; Farahani and Bayazidi, 2018). Excessive time outside, in the heat, and often near water, adds a physical challenge to sand work. Sand workers endure prolonged exposure to hot temperatures which can result in heat stroke (Rajesh and Naik, 2016) or skin diseases (Mohapatra, 2017). Sand work is also linked to a series of respiratory diseases, while the stagnant pools of water that sand mining creates are a breeding ground for malaria transmitting anopheles mosquitoes (Farahani and Bayazidi, 2018). Sand workers may likewise be exposed to silicosis or tuberculosis from working constantly in silica sand dust (Wazir, 2002; Singh and Tripathi, 2010), or they may be in direct contact with heavy metals or sand that may contain radioactive chemicals (zircon, titanium, radon) (Dung et al., 2014).

A third category of articles unpack how sand work can heighten conflict amongst households and within communities. The physical act of sand mining – particularly dredging which can take place day and night – has been described as a noise irritant by local populations exposed to sand dredging (Cho, 2006; Lamb et al., 2019). Sand mining activities also introduce conflict between fishers and mining operators who are competing for the same marine space (Kim, 2009). Local inequalities are further heightened when sand work is introduced or privileged over other livelihood activities: sand mining results in some people getting jobs, which foments resentment and anger between villagers (Farahani and Bayazidi, 2018). Such tensions further increase when people are being displaced through sand mining activities directly or because of how sand mining damages ecosystems and infrastructure such as roads and bridges (Farahani et al., 2018; Ramachandra et al., 2018). In areas with an active sand mafia presence, violence may emerge as various sand mafias compete with one another for sand and labour (Rege, 2016; Rege and Lavorgna, 2017).

The fourth category of papers illustrates how sand mining activities impact other types of work, often linked to ecological decline, some of which we detailed in section 3.2. Sand mining in river and delta estuaries impacts fishing practices, as aquatic species shift their fishing grounds away from sand mining sites (Kim, 2009; Lamb et al., 2019). Not all fishers have boats that can handle deeper waters or further trips from home, nor handle higher fuel expenditures. Sand mining-driven erosion and shifting delta ecosystems also impact riverbank farmers (Lamb et al., 2019), and farmers more generally (Pathania et al., 2017). Agriculture and fisheries are key livelihood activities for people living in rural areas throughout Asia, and negatively impacting either of these can reduce livelihood diversity and may increase poverty levels (Pathania and Singh, 2017; Farahani and Bayazidi, 2018). Sand mining leads to significant habitat loss, including habitat for migratory bird species which can be a source of coastal tourism (Rosyidie et al., 2018). Sand mining also ruins the aesthetics found in coastal areas, impacting communities, but also tourism (Rege and Lavorgna, 2017). Only one study in our review focused on how global flows of sand affect people making a living across multiple sites of sand extraction (Lamb et al., 2019).

3.4. Common grounds: weak sand governance

Weak sand governance transcends individual Asian countries, with authors documenting wide-scale illegal sand mining across China (Chen et al., 2006), Korea (Cho, 2006), India (Aarif et al., 2014; Tepjal et al., 2014), Iran (Farahani and Bayazidi, 2018), Malaysia (Ashraf et al., 2011), and Cambodia (Lamb et al., 2019). Sand governance is mentioned in relation to sand mining bans (in riverine, coastal, or inland ecosystems), or sand export bans, as have been seen in Cambodia (Lamb et al., 2019), China (Chen et al., 2006; Meng et al., 2018; Duan et al.,

2019) and India (Tejpal et al., 2014; Bisht and Gerber, 2017; Singh & Kumar 2018). These are ad hoc, geographically-constrained measures, that often displace sand mining efforts elsewhere. For example, Poyang Lake only emerged as the world's most important sand mining site after a sand mining ban was issued and enforced on the Yangtze River in 1998, in the wake of that year's massive floods (Zheng et al., 2016; Tian et al., 2017). Other authors highlight that sand mining bans lead to cascading impacts, including operators ignoring licencing limits or engaging in sand mining without a permit; this, in turn, reshapes sand prices and affects infrastructure schemes, real estate markets, and development priorities (Tejpal et al., 2014; Singh and Kumar 2018).

Indeed, the role of infrastructure development in driving sand mining cannot be understated in the Asian region. Insufficient sand legislation (Ashraf et al., 2011) and complex regulatory frameworks filled with loopholes (Cho, 2006) further contribute to weak sand governance. There is a level of opaqueness in terms of understanding how mining permits are given, volumes that can be extracted, how dredging licenses work, site restoration guidelines, etc. (Rege et al., 2017). None of the papers we examined looks at sand mining quotas or provides a picture of how particular government departments work with regards to sand mining. Nor is much known about sand mining companies. A challenge facing sand mining governance that does stand out, though, is coordination across levels of government bureaucracy, as conflict can emerge between various administrative units (Chen et al., 2006; Al-Awhadi et al., 2013), including those charged with environmental protection and those charged with resource development.

Although sand is cheap on a per tonne basis – sand from Myanmar and Cambodia can be less than USD 5 per metric tonne (Lamb et al., 2019) – extracting large volumes of sand can be profitable. For example, Duan et al. (2019) found that operating a sand dredger over a 10-h period in Lake Poyang could yield profits ranging between USD 3,000–5,000. Sand mining offers a few actors a lucrative livelihood, including sand mafias, local elites, and, in some cases, corrupt politicians (Chen et al., 2006; Rege and Lavorgna, 2017; Lamb et al., 2019). This is further seen in the mis-match of international trade data: between 2007 and 2016 Cambodia reported exporting nearly 3 million metric tonnes of sand to Singapore; in contrast, in the same period, Singapore reported importing over 80 million metric tonnes (UNCTAD, 2021). This gap hints at the leaky, opaque nature of sand supply chains, where a few actors may make huge profits.

There is also a criminal element to sand mining. For example, up to 90 percent of all sand mines and half of mineral production development likely operates through criminal activities in India (Bisht and Gerber, 2017). Actors exploit poor regulatory frameworks, but also gather support from various segments of the population including real estate developers and authorities (Tejpal et al., 2014; Rege, 2016; Rege and Lavorgna, 2017). In areas with an active sand mafia presence, intensive sand mining activities are perpetuated through violence, intimidation, and corrupting vast networks of government representatives and authorities (Chen et al., 2006; Feng et al., 2012; Rege, 2016; Rege and Lavorgna, 2017), hinting at broader societal consequences. Bisht and Gerber (2017: 549) refer to these phenomena as 'predatory extractivism', namely a form of exploitation where local populations are left to cope with the social and environmental externalities of extractive industries such as sand mining (see also Al-Awadhi et al., 2013).

Even with this grim sand governance outlook, the literature points to potential sand governance solutions, particularly in the area of environmental regulation. Some recommendations directly speak to the sand governance challenges highlighted above. These include: enhancing legal frameworks in China (Meng et al., 2018), India (Rege and Lavorgna, 2017) and Malaysia (Ashraf et al., 2011); enforcing sand mining permits and using satellite imagery to detect illegal sand mining operations (Chaussard and Kerosky, 2016; Duan et al., 2019); conducting frequent permit assessments (Al-Awadhi et al., 2013; Singh and Kumar 2018); allocating longer leases to dredgers to encourage sustainable resource management (Rege, 2016); or fostering coordination between

administrative units, including within China and in Oman (Chen et al., 2006; Al-Awadhi et al., 2013). Similarly, the need for more transparent administrative and policy-making processes that encourage local participation are noted (Rege, 2016; Farahani and Bayazidi, 2018). In this sense, sand governance faces challenges similar to other supply chains characterized by illicit supply networks that are difficult to govern, such as the wildlife trade (Magliocca et al., 2021).

One recommendation consistently noted is for countries to follow their own environmental impact assessment (EIA) policies. Pleas for systematic EIAs have been made for China (de Leuw et al., 2010; Meng et al., 2018), India (John et al., 2017); Cambodia (Lamb et al., 2019) and Iran (Farahani and Bayazidi, 2018). Scholars likewise highlight the need to identify and select suitable sites for sand extraction (Ashraf et al., 2011; Al-Awadhi et al., 2013; Farahani and Bayazidi, 2018), to minimize impacts through exploiting optimal sand quantities, and accounting for biotic and climatic seasonality (Shao et al., 2012; Tejpal et al., 2014; Rehitha et al., 2017; Saleh et al., 2017; Ramachandra et al., 2018; Sadeghi et al., 2018). A win-win scenario for dredging in hydroelectric reservoirs is proposed to also help prevent premature reservoir aging (Chen et al., 2006). Digging depth and mine area could be limited in various mining environments (Ashraf et al., 2011; Ghosh et al., 2016; Meng et al., 2018), and decommissioned mines could be restored with topsoil left aside prior to dredging in order to rehabilitate their biotic and abiotic integrity (Ashraf et al., 2011; Al-Awadhi et al., 2013; Singh and Kumar 2018). Such restoration plans could be financed through taxes or royalties aimed at internalizing the environmental externalities that sand mining creates (Kim, 2009; John et al., 2017; Singh, 2017).

Scholars aptly note that further research is needed to understand the scale of sand mining across Asia, its impacts and externalities, and the extent to which extraction aligns with ecological carrying capacities (Kim et al., 2008; Ashraf et al., 2011; Sreebha and Padamal, 2011; Sreekumari et al., 2016; John et al., 2017; Meng et al., 2018), recommendations which the UNEP (2019) echoes. Under-reporting and/or a lack of data on market demand and extraction rates notably stand out as major anomalies given the size and strategic importance of the sand mining sector (Li et al., 2014; Bisht and Gerber, 2017; Farahani and Bayazidi, 2018). National accounting for sand mining is patchy at best. The literature also includes calls for research to promote 'sustainable' or holistic sand resource development models, though these often remain vague and poorly substantiated (Cho, 2006; Sheik and Chandrasekar, 2011; Tejpal et al., 2014; Farahani and Bayazidi, 2018; Ramachandra et al., 2018). A few papers, particularly those focusing on India (but not only), call for greater sand activism (Ashraf et al., 2011; Tejpal et al., 2014; Bisht and Gerber, 2017; John et al., 2017). That said, the criminal networks involved in this sector, together with the authoritarian nature of many Asian governments make campaigning risky, as the jailing of Mother Nature activists in Cambodia and suppression of activists in India demonstrate (Larson, 2018; Beiser, 2018).

4. Discussion and conclusion

The demand for sand will continue: at a global level, the extraction of non-metallic minerals, which include sand, gravel and limestone, is expected to nearly double from 44 Gt to 86 Gt between 2017 and 2060 (OECD, 2019). Although there are debates about the extent of sand scarcity overall (see Torres et al., 2021), regional sand scarcity is an emerging issue (i.e., physical scarcity ensues when demand exceeds physical availability and regeneration rates). China and India have the highest sand demand globally, with Southeast Asian countries forecasted to experience a rapid increase in sand demand over the next 15 years (Ioannidou et al., 2020). This demand will result in risks to the sand supply system across the Asian region, further compounding current socio-environmental challenges.

This systematic scoping review highlights the serious ecological impacts of unabated sand mining in Asia, including its biotic and abiotic consequences. Sand mining triggers significant erosional processes:

lower sediment discharge acts as a sediment sink for riverbed and riverside aggregates (Jordan et al., 2019) impacting homes and livelihoods across river and delta communities. Moreover, massive silt removal likely disrupts land fertility regimes along Asia's rivers – the deltas of which form the region's rice granaries – and sand mining likely contributes towards the spread of invasive species (Torres et al., 2021). These two issues require further research. Insights from other regions, including sand mining in other BRIC countries, can add further ecological understanding including the ecosystem service impacts of sand mining from a longitudinal perspective (see Ferrer et al., 2021).

The biological changes emerging from sand mining are relatively well documented for the Mekong Delta and Pearl River Delta; however, many other Asian regions remain vastly understudied. Wetlands, coastal and numerous riverine waters as well as marine environments are being altered or lost in favour of a landscape that can handle the built form. Some species adapt, some species can recover after mining sites are decommissioned, others cannot (Torres et al., 2017). Only a few studies are able to quantify the economic costs of such alteration, including costs associated with erosion or species loss, but also wetland infill or built-up urban spaces. The pace of urbanization – Asian cities currently add more inhabitants than anywhere else – requires a massive, steady supply of cheap sand.

Asian scholars are at the forefront of the Asian-based sand literature, investigating sand ecologies to a far greater extent than sand livelihoods or other social dimensions. This focus can be explained, in part, by the presence of authoritarian governments that might discourage social scientists from probing into the politics of sand mining. For instance, Chinese and foreign social scientists have long been cautious when approaching 'sensitive' (*mingan*) research topics, including those pertaining to hydropower expansion (Rousseau and Habich-Sobiegalla, 2021). The recent intensification of academic censorship strategies by the Chinese government (Wong and Kwong, 2019) might further deter scholars to investigate the contentious politics and economics of sand mining. We also see this with the crackdown of liberal media in Cambodia (Beban et al., 2019), and this is likely true for other Asian countries with a poor track record on academic freedom. But, also, sand mining can be a violent activity involving criminal networks and illicit trade flows (Magliocca et al., 2021). This makes sand difficult to study, and further contributes to the challenges of understanding the social sustainability dimensions involved with sand.

Even so, what has been studied hints at the livelihood challenges that emerge from sand work, and regional differentiation. For example, across South Asia there is a reliance on workers to shovel and load sand, in contrast to Southeast or East Asia where sand work tends to be mechanized (dredgers, sand pumpers, barges) and fewer workers are needed. Sand jobs can involve coercion, violence, or forms of bonded or peonage labour (Singh and Tripathi, 2010; Vice News, 2021). Greater attention is needed to understand sand labour regimes and the extent to which mechanization levels and regional labour dynamics – including both organized and spontaneous work migration – affect working conditions in the sand industry. Occupational health and safety is another labour challenge: initial findings point to a series of health risks for those engaged in sand work including prolonged exposure to heat, respiratory and ophthalmic challenges, and the incidence of various skin conditions.

Contrary to many European countries that have prohibited instream sand mining and rely upon more expensive marine resources, as per the Torres et al. (2021) 'idealized' evolution of sand supply networks, Asian countries still predominantly exploit cheaper and more easily accessible riverine and coastal resources, with more visible impacts upon both ecosystems and livelihoods. Local populations – rural and urban alike – are typically left to cope with the socio-ecological consequences from sand mining on their own. Sand governance in Asia remains a challenge: there continues to be no global governance body for sand, there are gaps in international trade data (i.e., the mismatch of import-export statistics), and domestic production data are understated (UNEP, 2019). A great extent of sand across Asia is used domestically or in the region, and

the nature of the sand system (Torres et al., 2021) – from deposits to sand suppliers to sand demand – is not well understood. Criminality throughout the sand supply chain will continue to be an issue as sand supply shrinks (Rege, 2016; Rege and Lavorgna, 2017), and it is worth thinking about sand governance in the context of illicit supply networks (Magliocca et al., 2021) and precarious labour arrangements (Mahadevan, 2019; Marschke et al., 2021).

We note several other gaps in the literature we reviewed. Firstly, the literature focuses on China or India - 75 percent of the articles center on these two countries. Granted both China and India are major players when it comes to sand use (Ioannidou et al., 2020) and rapid urbanization (UN-Habitat, 2021), but this geographic concentration leaves important knowledge gaps into how sand mining and its impacts unfold elsewhere in the Asian region. Such geographic imbalances are also seen at a national scale. For example, the China-based scholarship almost exclusively probes the environmental impacts of sand mining in eastern China along the Yangtze and Yellow rivers, around Lake Poyang and in the Pearl River Delta, even as sand mining occurs in many more locations. In India, most research centers on coastal areas, with the southern states of Kerala and Tamil Nadu receiving much more attention than other territories. This potentially creates an undesirable path dependency scenario in characterizing sand research in Asia.

Secondly, environmental impacts related to sand mining do not occur in isolation. Many papers do not grasp the complexity of environmental change facing riverine, delta and coastal ecosystems, particularly anthropogenic change. The Asian region is experiencing rapid urbanization with major cities expanding on the edge of rivers or deltas (i.e., the Pearl and Yangtze river delta regions of China, cities along the Mekong Delta, but also the river banks of Bangkok and Calcutta to name a few), and the impacts of sand mining there add to those from other anthropogenic activities that include land infill, reforestation and hydropower expansion (see Hackney et al., 2020). Sand extraction and demand cannot be isolated from these impacts, nor from climate change. Liu et al. (2018) found that anthropological activities now drive greater sediment regime changes in the Pearl River system than any other driver of change. The importance of sand in mitigating and adapting to climate change impacts, in particular sea level rise which will hit the region hard, is absent in the literature (Torres et al., 2021).

Thirdly, when sand governance is mentioned, it is at a local or national scale. While this is important, and reflects the scale that particular papers focused on, transboundary solutions for sand governance are also critical to consider (Lamb and Fung, 2022). Sand exporting communities experiencing the environmental and livelihood impacts of sand mining may be far away from sand receiving areas, or importing countries when sand crosses national borders. Yet, regional sand demand has multiple, often cascading, consequences that accumulate in the sand system over time. For example, as Singapore's territory expanded by a fifth between 1965 and 2015, the city state's main sand suppliers began with Indonesia and Malaysia, switched to Cambodia and Vietnam, and now relies significantly on India and Turkey (UNCTAD, 2021). One by one, in response to environmental impacts such as erosion or fish stock decline and to social tensions in sending regions, each country banned coastal sand exports (albeit sometimes temporarily). Regional and transboundary sand governance mechanisms and strategies stand out as solutions to halt this ever-shifting cycle.

Sand has received far less attention than other natural resources – including water, forest, land, fish, and so on. The emerging sand scholarship aims at grasping the specific and 'granular' geographies that sand mining brings about (Jamieson, 2021a), reflects on the global sand supply system (Torres et al., 2021), and places sand in discussion with other illicit commodities (Magliocca et al., 2021). These are important advances. What our literature review further highlights is how such thinking must now grapple with the specificities across the Asian region, with all its variations, and far greater attention needs to be paid to the people involved throughout sand supply chains. This is not only because Asia has used more sand than any other region; but also because

urbanization and sand demand scenarios make it clear that this will remain so for the foreseeable future.

Author statement

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